



An Integrated Energy Conversion and Control Framework for Grid-Connected Photovoltaic Systems

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Abstract: This study presents an integrated strategy for both energy conversion and control in a grid-connected photovoltaic (PV) system. The goal is to ensure smooth coordination between power extraction, direct-current (DC) link voltage regulation, grid synchronization, and control of power exchange. The system configuration includes a PV array, a Boost converter controlled by a hybrid maximum power point tracking (MPPT) algorithm combining Perturb and Observe (P&O) with the Whale Optimization Algorithm (WOA), DC-link stage, and a three-phase inverter. Grid synchronization is achieved using a synchronous reference frame phase-locked loop (SRF-PLL), while active and reactive power are managed through indirect vector control in the synchronous dq reference frame. The model was developed and tested in MATLAB/Simulink 2022b under two conditions: a steady-state scenario at an irradiance of 1000 W/m² and a temperature of 25 °C, and a dynamic scenario with varying irradiance while maintaining constant temperature. In steady-state operation, the PV system delivers around 100 kW, the DC-link voltage is stabilized near 600 V, and the grid voltages remain balanced with peak values close to ±340 V. The injected currents also show good quality with low distortion. When subjected to changes in irradiance, the system responds effectively, with the extracted power closely tracking the variations. At the same time, the DC-link voltage remains stable, the three-phase voltage balance is preserved, and the quality of the injected current is maintained. Overall, the results confirm that the proposed method offers a reliable and well-coordinated solution for integrating energy conversion and control in grid-connected PV systems, delivering strong performance in both steady-state and dynamic operating conditions.

Keywords: Grid-connected photovoltaic system; Maximum power point tracking; DC-link voltage control; Synchronous reference frame phase-locked loop; Synchronous dq-frame vector control

1 Introduction

1.1 Context and Motivation

The global shift toward cleaner energy is driving a rapid expansion of renewable power sources in modern electrical grids. Among these technologies, photovoltaic (PV) systems have become especially important due to their flexibility, falling costs, and ability to scale easily—from small residential setups to large utility-scale plants [1]. As a result, grid-connected PV systems are now central to reducing carbon emissions and reshaping traditional power networks into more decentralized, adaptive, and intelligent infrastructures. Despite these advantages, the widespread adoption of PV systems brings several technical challenges. Solar energy is inherently intermittent and cannot be dispatched on demand, which complicates its integration into the grid. Therefore, PV systems must go beyond simple energy conversion and meet strict requirements related to system stability, synchronization with the grid reliability, and power quality [2]. In real-world operation, this means ensuring maximum energy extraction while also maintaining stable electrical performance, minimizing harmonics, and effectively controlling both active and reactive power. Given these constraints, developing a conversion and control strategy that can simultaneously deliver high efficiency, strong dynamic performance, and compliance with grid standards has become a critical issue.

Addressing this challenge requires a holistic design approach, where energy conversion, control mechanisms, and synchronization are not treated separately but as closely linked elements of a single, coordinated system [3].

1.2 Scientific Problem

A grid-connected PV system can be viewed as a highly integrated multi-stage architecture where electrical behavior and control mechanisms are tightly interconnected. Such a system generally consists of a PV array, a maximum power point tracking (MPPT) unit, a direct-current-to-direct-current (DC–DC) boost converter, a DC-link capacitor, a voltage source inverter, an output filter, and a synchronization mechanism based on a phase-locked loop (PLL). As a result, the overall efficiency and reliability of the system rely not only on the individual performance of each component, but also on their effective dynamic interaction and coordination. This strong coupling introduces several important operational challenges. The controller must rapidly and accurately extract maximum power despite continuous changes in solar irradiance and temperature conditions. At the same time, maintaining DC-link voltage stability is crucial for ensuring uninterrupted inverter functionality and efficient power transfer. The system is also required to regulate active and reactive power injection in accordance with grid requirements. Furthermore, minimizing the total harmonic distortion (THD) of the injected current is necessary to maintain acceptable power quality and reduce adverse effects on the utility network. In addition, the overall configuration must demonstrate strong resilience against disturbances, especially sudden variations in irradiance that directly influence both the PV source characteristics and the associated control response.

Therefore, the primary difficulty is not limited to designing independent controllers for individual subsystems, but rather lies in achieving coordinated control of a nonlinear and strongly coupled system operating under continuously changing environmental conditions.

1.3 Research Gap

Although grid-connected PV systems have been widely investigated, most existing studies examine the different stages of the conversion process independently. Several works concentrate individually on MPPT strategies, DC–DC converter operation, inverter control, or grid synchronization. While such an approach is valuable for evaluating subsystem-level performance, it does not adequately represent the overall dynamic behavior of PV systems under practical operating conditions.

More specifically, comprehensive studies connecting PV array behavior, DC-link voltage control, PLL-based synchronization, and grid-side power quality remain limited. These components are strongly interconnected: disturbances originating from the PV source may affect injected grid currents, whereas poor DC-link regulation or synchronization can negatively influence both stability and energy efficiency. This limitation emphasizes the necessity for coordinated system-level control strategies capable of simultaneously achieving optimal power extraction, stable operation, and compliance with grid requirements.

1.4 Contributions

Within this framework, the present study introduces a unified approach for control and energy management in grid-connected PV systems. Initially, a detailed model of the complete PV conversion chain is established, including the PV array, Boost converter, DC-link stage inverter, coupling filter, and grid synchronization unit, thereby providing a consistent description of the interactions among system components. Next, a coordinated control strategy is proposed by integrating MPPT operation, DC-link voltage stabilization [4], PLL-based synchronization and active/reactive power regulation at the grid interface allowing synchronized control of both energy extraction and power injection. The dynamic response of the system under varying irradiance conditions is then investigated to evaluate controller robustness and its capability to preserve stable electrical and energy performance during environmental fluctuations [5].

Lastly, the performance of the injected power is examined through grid-current behavior, power transfer analysis, and THD assessment demonstrate that the proposed framework simultaneously satisfies stability, efficiency, and power quality objectives.

2 Literature Review

2.1 Maximum Power Point Tracking Control Review

Table 1 provides a selective and analytical review of recent contributions to MPPT techniques for PV systems. It compares the control approaches, evaluation contexts, methodological contributions, identified limitations, and relevance of these techniques to grid-connected PV architectures.

A review of recent studies indicates a noticeable transition from traditional MPPT techniques toward intelligent, hybrid, and metaheuristic-based approaches. This evolution is mainly motivated by the demand for faster convergence, more accurate global maximum power point (GMPP) detection, and improved robustness under rapidly changing environmental conditions and partial shading scenarios. Conventional MPPT methods continue to be appealing

because of their simple structure and ease of implementation; however, their effectiveness decreases significantly in the presence of multiple power peaks.

Table 1. Literature review of MPPT techniques applied to grid-connected PV systems

Ref.	MPPT Control Type	Study Conditions	Main Contributions	Limitations	Relevance to This Study
[2]	Intelligent (ML)	Variable irradiance/temperature	Introduces a lightweight decision-tree-based AI approach.	Less validated than ANN/FLC methods.	Relevant for comparing lightweight AI and hybrid approaches.
[6]	Conventional + Intelligent	Boost converter, same PV system, MATLAB/Simulink	Provides a fair comparison of FLC, InC, and P&O in terms of speed, accuracy, and stability.	Limited to simulation validation.	Useful as a baseline comparison.
[7]	Improved conventional	Variable irradiance, experimental validation	Reduces ripple and improves efficiency with both simulation and experimental validation.	Limited comparison scope.	Highly relevant for practical discussion.
[8]	Hybrid intelligent	Dynamic conditions	Combines fuzzy logic with hybrid techniques to improve MPPT tracking.	Mainly simulation-based validation.	Relevant for hybrid FLC discussion.
[9]	Comparative study	Uniform irradiance & partial shading	Compares recent MPPT families in terms of speed, complexity, and robustness.	Focused mainly on MPPT function.	Useful for comparative conclusions.
[10]	Hybrid intelligent + metaheuristic	Grid-connected, variable conditions	Integrates intelligent MPPT with grid-connected PV using ANFIS-PSO optimization.	High implementation complexity.	Highly relevant to grid-connected PV context.
[11]	Conventional vs AI	General/simulation	Compares classical and AI approaches under unified criteria.	Not focused on full conversion chain.	Supports comparative argumentation.
[12]	Hybrid metaheuristic	Variable weather conditions	Improves global search through GA-PSO hybridization.	Increased computational cost.	Relevant for metaheuristic discussion.
[13]	Review	General	Provides a recent synthesis of MPPT approaches.	Limited detailed results in some sections.	Useful as a recent review reference.

[14]	Intelligent (FLC)	MATLAB simulation/Boost	Achieves 97% efficiency with improved stability and reduced fluctuations.	Limited experimental validation.	Supports FLC-based approaches.
[15]	Hybrid intelligent	Variable irradiance/temperature	Combines LSTM-ANN and fuzzy logic for dynamic MPPT.	Requires training data and tuning.	Relevant for advanced AI approaches.
[16]	Intelligent (ANN)	Grid-connected PV-wind-battery system	Applies ANN-based MPPT trained using the Levenberg-Marquardt algorithm.	Higher implementation complexity.	Relevant for ANN-based MPPT.
[17]	Hybrid metaheuristic + FLC	Grid-connected PV-battery system	Combines global search and fast response via BAT-Fuzzy.	Focuses on PV-battery systems.	Useful for hybrid control discussion.
[18]	AI review	PV analysis/control	Highlights the role of AI in PV control and evaluation.	Not focused solely on MPPT.	Useful for AI subsection.
[19]	Optimization/hybrid review	General	Emphasizes trends in optimization, hybridization, and AI methods.	Broad scope beyond MPPT.	Strong support for hybrid approach discussion.
[20]	Hybrid intelligent + metaheuristic	Grid-connected PV	Targets optimized MPPT for grid-connected systems.	Focuses mainly on MPPT performance.	Highly relevant for final discussion.

Note: MPPT = maximum power point tracking; ML = machine learning; AI = artificial intelligence; ANN = artificial neural network; FLC = fuzzy logic controller; InC = incremental conductance; P&O = Perturb and Observe; PV = photovoltaic; ANFIS = adaptive neuro-fuzzy inference system; PSO = particle swarm optimization; GA = genetic algorithm; LSTM = long short-term memory; BAT = Bat Algorithm.

In contrast, hybrid approaches provide better adaptability and enhanced global search performance, although they generally introduce higher computational and structural complexity [6]. Despite these advances, the literature still lacks sufficiently integrated studies addressing the interactions among MPPT algorithms, DC–DC converters, DC-link regulation, inverter control, and grid-side power quality in interconnected PV systems. The adoption of the hybrid Perturb and Observe (P&O)–Whale Optimization Algorithm (WOA) technique is motivated by the need to balance implementation simplicity, computational efficiency, and global optimization capability. The Perturb and Observe algorithm remains suitable for real-time applications due to its lightweight design, straightforward implementation through duty-cycle control of the Boost converter, and rapid local tracking near the operating optimum. However, under partial shading conditions [7], where the power–voltage (P–V) curve becomes multimodal, the method may converge to a local rather than global maximum. Incorporating the WOA addresses this limitation by introducing an effective global search mechanism capable of locating the GMPP in complex non-convex power profiles. Compared with more computationally intensive hybrid methods, the P&O–WOA approach offers an efficient compromise between local accuracy, global exploration, and practical integration within grid-connected PV control systems [8].

To make the choice of the search mechanism explicit, the reviewer’s three candidate motivations for the P&O–WOA hybridization—faster tracking, lower computational burden, and better performance under complex operating conditions—were investigated through a dedicated, harmonized benchmarking study conducted on the same PV–boost architecture. Classical P&O was compared against four bio-inspired hybridizations, namely P&O–WOA, P&O–GWO (Grey Wolf Optimizer), P&O–BAT (Bat Algorithm), and P&O–SFLA, under a strictly identical computational budget (population size $N = 10$, maximum $\text{Iter}_{\max} = 20$ iterations per MPPT decision, i.e. 200 fitness evaluations per control cycle) and an identical 1 ms control-loop sampling time for all variants. Because the computational budget is harmonized across all hybrids, the comparison isolates the effect of the search-mechanism itself rather than differences in allotted computation.

Table 2 reports the results obtained under Standard Test Conditions (1000 W/m², 25 °C), and Table 3 reports the re-tracking time and qualitative lock-in risk under a stepwise variable-irradiance profile (1000→700→400→600→800 W/m²).

Table 2. Comparative MPPT performance of P&O and its hybrids under Standard Test Conditions (1000 W/m², 25 °C)

Method	t_r (ms)	t_s (ms)	IAE	ITAE	ISE	η_{MPPT} (%)
P&O (baseline)	44.60	64.00	252.50	85.12	2.25	97.37
P&O-WOA	31.46	48.00	238.30	84.55	2.02	98.44
P&O-SFLA	32.30	49.00	239.60	84.59	2.04	98.38
P&O-BAT	33.00	50.00	241.30	84.75	2.07	98.38
P&O-GWO	35.90	53.00	244.10	84.86	2.13	98.40

Note: MPPT = maximum power point tracking; η_{MPPT} = MPPT efficiency; P&O = Perturb and Observe; t_r = rise time; t_s = settling time; IAE = integral absolute error; ITAE = integral time absolute error; ISE = integral squared error; WOA = Whale Optimization Algorithm; SFLA = Shuffled Frog-Leaping Algorithm; BAT = Bat Algorithm; GWO = Grey Wolf Optimizer.

Table 3. Re-tracking performance comparison

Method	Re-Tracking Time (ms)	Lock-in Risk
P&O (baseline)	65.0	High
P&O-WOA	32.5	Low
P&O-SFLA	32.5	Very low
P&O-GWO	50.0	Low
P&O-BAT	52.5	Low

Note: P&O = Perturb and Observe; WOA = Whale Optimization Algorithm; SFLA = Shuffled Frog-Leaping Algorithm; BAT = Bat Algorithm; GWO = Grey Wolf Optimizer.

Under this harmonized protocol, P&O-WOA achieved the fastest transient response of all four hybrids (t_r = 31.46 ms , a 29.5% reduction relative to classical P&O ; t_s reduced by 25%), the largest reduction in squared tracking error (ISE: -10.48%), and the highest steady-state MPPT efficiency (98.44%), while its re-tracking time under variable irradiance was reduced by roughly 50% relative to classical P&O and its lock-in risk under single-peak transitions was rated Low, on par with SFLA. Since all four hybrids operate under the identical computational budget ($N \times \text{Iter}_{\text{max}} = 200$ evaluations/cycle), and WOA's update rule-encircling, spiral and search mechanisms governed by a single linearly-decreasing convergence parameter a and a spiral constant b - is no more demanding than GWO's single-parameter update, and considerably simpler than BAT's multi-parameter pulse-rate/loudness scheme or SFLA's memplex-shuffling structure, the P&O-WOA combination was selected primarily because it improves tracking speed and robustness against local optima under multimodal/variable-irradiance conditions, and not because it reduces computational burden relative to the other hybrids evaluated-all four impose a comparable per-cycle load.

2.2 Literature Review of Synchronous Phase-Locked Loop Techniques

Table 4 provides a critical and structured review of recent studies on PLL-based synchronization techniques in grid-connected PV systems. It summarizes the different PLL types employed, the grid conditions investigated, the main contributions reported, the identified limitations, and their relevance to grid-connected power conversion architectures.

A thorough examination of existing research highlights the essential role of PLL-based synchronization in ensuring stability, control precision, and power injection quality in grid-connected PV systems. This importance arises from its direct impact on dq frame decoupling, active and reactive power regulation, and the harmonic characteristics of the injected currents. Advanced synchronization methods, including DSOGI-PLL structures, demonstrate improved performance under weak, unbalanced, or highly distorted grid conditions. However, their advantages become particularly significant only in severely disturbed operating environments. In the context of the present work, which assumes a balanced or moderately disturbed three-phase grid, the synchronous reference frame (SRF)-PLL is considered the most appropriate solution. This choice is motivated by its straightforward implementation, effective compatibility with synchronous dq-frame vector control, and reliable estimation of grid phase and frequency without introducing unnecessary control complexity. Consequently, the selection of the SRF-PLL is driven not by the pursuit of sophisticated synchronization structures, but rather by the objective of achieving an efficient balance between synchronization performance, dynamic robustness, and injected power quality.

Table 4. Literature review of synchronous PLL techniques

Ref.	PLL/ Synchronization Type	Grid Conditions Studied	Main Contributions	Limitations	Relevance to This Study
[21]	PLL-stability review	Weak grid	Analyzes the impact of PLL bandwidth and its interactions with other control loops.	Strongly focused on weak-grid conditions.	Highly useful for linking PLL behavior to overall system stability.
[22]	DSOGI-PLL	Distorted grid, PV system	Applies DSOGI-PLL to a grid-connected PV structure based on a Z-source inverter.	Specific topology.	Highly relevant for PV applications.
[23]	Hybrid GI-PLL + LADRC	Non-ideal grid conditions	Provides strong DC component rejection and fast dynamic response.	More complex structure.	Useful for advanced PLL variants.
[24]	PLL stability analysis	Weak grid, multi-converter interactions	Shows that grid structure directly affects PLL stability.	Conference paper.	Highly useful from a system-level perspective.
[25]	Small-signal DSOGI-PLL	Distorted grid	Provides a detailed analytical basis for the frequency behavior of the DSOGI-PLL.	Highly theoretical.	Useful for the modeling section.
[26]	General review	Healthy and disturbed grids, grid-tied PV	A key review of synchronization techniques for grid-connected PV systems.	More focused on synchronization than on overall control.	A strong recent reference.
[27]	DSOGI-based PLL	Weak grid, harmonics, sag, unbalance	Compares several DSOGI-PLL variants under realistic disturbances.	Strongly focused on DSOGI.	Highly useful for justifying a robust PLL choice.
[28]	Improved DSOGI-PLL	Disturbed grid	Improves dynamic response with a simple approach without degrading disturbance rejection.	Mostly limited to the PLL loop itself.	Good technical support.
[29]	Modified SRF-PLL	Highly distorted distribution grids	Proposes a robust PLL for synchronization and frequency monitoring in highly distorted distribution grids.	More oriented toward monitoring and distribution grids.	Highly useful for implementation discussion.

[30]	Comparative study (PSRF-PLL, SOGI-PLL, DSOGI-PLL, E-PLL, IPT-PLL)	Sag, swell, unbalance, harmonics	Compares several PLL families under realistic scenarios for distributed generation sources.	Limited direct connection with MPPT/DC-link issues.	A very strong comparative reference.
[31]	Improved DSOGI-PLL	Non-ideal grid conditions	Compares SRF-PLL, conventional DSOGI, and improved DSOGI under non-ideal conditions.	Mainly focused on phase-estimation accuracy.	Very good technical support.
[32]	Review/nonlinear stability analysis	Renewable-dominated systems	Links synchronization stability to the nonlinear dynamics of grid-connected converters.	Highly theoretical.	Useful for strengthening the scientific discussion.

Note: PLL = phase-locked loop; E-PLL = enhanced phase-locked loop; IPT = improved park transformation; SOGI = second-order generalized integrator; DSOGI = dual second-order generalized integrator; GI = generalized integrator; LADRC = linear active disturbance rejection control; SRF = synchronous reference frame; PSRF = positive sequence reference frame; dq = direct-quadrature reference frame; PV = photovoltaic; DC = direct current; MPPT = maximum power point tracking.

2.3 Active and Reactive Power Control in the Synchronous dq Frame

Table 5 provides a critical overview of recent studies on control laws for grid-connected PV inverters. It highlights the relationships among control strategies, reference frames, control objectives, methodological contributions, and identified limitations, while assessing their applicability within a complete grid-connected PV architecture.

Table 5. Literature review of P/Q control techniques applied to grid-connected photovoltaic systems

Ref.	Control Type	Control Frame	Main Objective	Key Contributions	Limitations	Relevance to This Study
[33]	General review	Various	Overview of inverter topologies and control strategies	Provides a recent and structured overview of topologies and controls	Broad scope, not focused on full PV chain	Framing reference
[34]	Control + AI review	dq, various	State-of-the-art of control methods and AI in PV inverters	Highlights AI trends and conventional vs AI comparison	Very recent; needs complement with established works	Forward-looking reference
[35]	Review	dq, various	Link grid codes, topologies, and control techniques	Strong on grid codes, architectures, and control methods	Limited explicit link with MPPT/DC-link	Major Q1 reference
[36]	PI/DC-link loop	dq/AC + DC-link	Reduce DC-link voltage fluctuations	Explicitly links DC-link control and grid-side performance	Single-phase system	Highly relevant for DC-link control
[37]	MPC + sliding mode	Nonlinear	Improve robustness and disturbance rejection	Combines nonlinear MPC and sliding mode control	High implementation complexity	Useful for advanced alternatives

[38]	Multi-resonant	$\alpha\beta$ /stationary	Grid injection and compensation under disturbances	Demonstrates multifunctional inverter capabilities	Specific case study	Relevant for power quality discussion
[39]	MPC	dq	Improve stability considering PLL dynamics	Explicit integration of PLL dynamics into current control	Focused on weak-grid conditions	Highly relevant for control-PLL interaction
[40]	MPC review	dq, various	Survey MPC approaches in PV systems	Strong synthesis of MPC control families	Focused on a single control family	Useful for MPC subsection
[41]	Adaptive control review	dq, various	Summarize adaptive control methods	Strong focus on robustness and uncertainty handling	Less focused on full system architecture	Relevant for future perspectives
[42]	AI: RL vs ANN	Various	Compare AI methods for power quality	Compares RL and ANN in grid-connected PV systems	Practical robustness still uncertain	Useful for AI extension
[43]	Reactive power control	dq	Evaluate reactive power compensation strategies	Links inverter control with grid support	Focused on reactive power only	Relevant for P/Q control section
[44]	Advanced control (unbalanced grid)	dq	Improve stability under unbalanced conditions	Addresses DC-link oscillations under grid disturbances	Specific disturbed case	Useful for robustness discussion
[45]	Output-feedback/observer	dq	Sensorless control with power smoothing	Reduces sensor requirements and smooths power	Specific topology, single-phase	Complementary reference
[46]	Current control review	abc/dq	Review current control strategies	Strong technical support for AC-side control	Not PV-specific	Useful technical reference

Note: P/Q = active and reactive power; MPPT = maximum power point tracking; PI = proportional-integral; MPC = model predictive control; PLL = phase-locked loop; RL = reinforcement learning; ANN = artificial neural network; abc = three-phase stationary reference frame; dq = direct-quadrature reference frame; $\alpha\beta$ = alpha-beta stationary.

Recent studies reveal that, although numerous control techniques have been developed for grid-connected PV inverters, control approaches based on the synchronous dq reference frame continue to provide one of the most reliable solutions for satisfying both internal system regulation and grid-side operational requirements. In this framework, active and reactive power control using $\frac{i_d}{i_q}$ current regulation has proven to be especially efficient. By synchronizing the reference frame with the grid voltage, active power can be naturally controlled through the d-axis component, while reactive power is linked to the q-axis component. This arrangement allows the DC-link voltage to be effectively regulated via i_d , whereas i_q can be dedicated either to reactive power compensation or to maintaining unity power factor operation.

Compared with more sophisticated control schemes that often require greater computational effort and implementation complexity, this method provides a practical balance between simplicity, clear physical interpretation, dynamic stability, and ease of integration within a unified framework involving MPPT, PLL synchronization, DC-link regulation, and grid power injection.

3 Materials and Methods

3.1 Integrated Architecture of the Grid-Connected Photovoltaic System

The proposed grid-connected PV conversion system, shown in Figure 1 and parameterized as summarized in Table 6, consists of a PV array, a DC-DC Boost converter controlled by an MPPT algorithm, a DC-link stage, a

three-phase inverter equipped with an output filter, an SRF-PLL synchronization module, synchronous dq-frame vector control, and dedicated monitoring and performance-evaluation modules.

The PV generator operates under variable solar irradiance and temperature conditions, while a hybrid P&O-WOA MPPT strategy controls the Boost converter to maximize the extracted PV power. The DC-link stage acts as an intermediate energy buffer and maintains the DC-link voltage at its reference value. The three-phase inverter subsequently converts the generated DC power into AC power for injection into the utility grid, while the output filter attenuates switching harmonics and improves the quality of the injected current. Grid synchronization is achieved using the SRF-PLL, whereas grid-side power exchange is managed through cascaded vector control in the synchronous dq reference frame. Within this control structure, the d-axis current component, i_d , regulates the active power and contributes to DC-link voltage stabilization, while the q-axis current component, i_q , controls the reactive power. In the present configuration, the reference value i_q is set to zero to ensure operation at approximately unity power factor. In addition, power-measurement and THD-analysis modules are incorporated to evaluate the overall system performance, transient response, and quality of the power delivered to the grid.

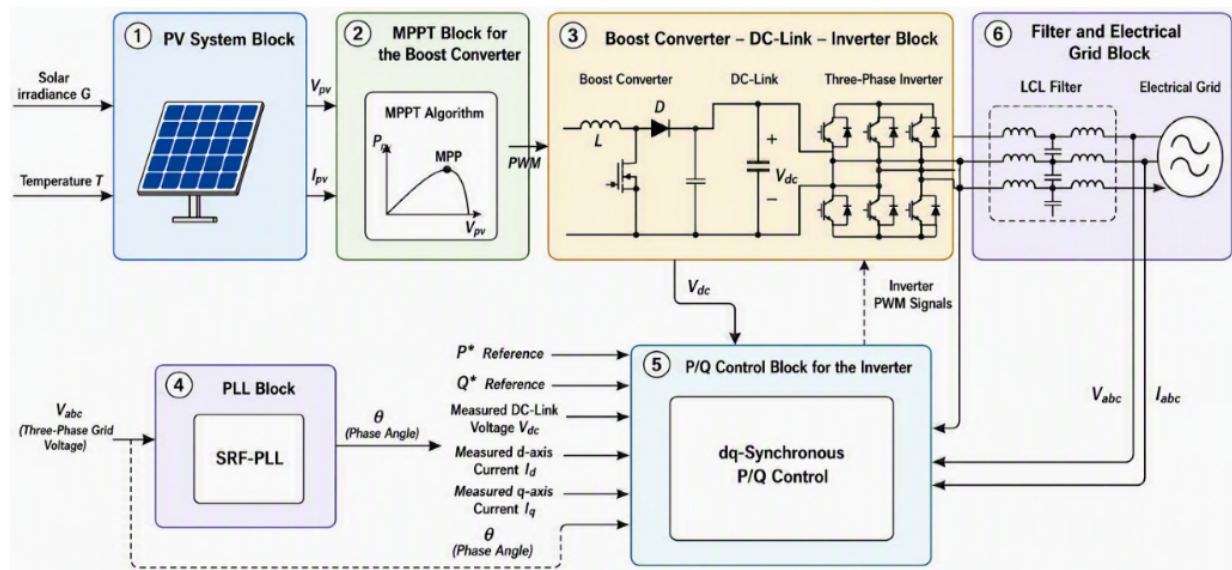


Figure 1. Grid-connected PV energy conversion chain

Note: PV = photovoltaic; MPPT = maximum power point tracking; DC = direct-current; SRF-PLL = synchronous reference frame phase-locked loop; P/Q = active and reactive power.

Table 6. Electrical and control parameters of the grid-connected PV conversion system

Parameter	Symbol	Value	Unit
Number of parallel strings	N_p	18.00	—
Series-connected modules per string	N_s	9.00	—
Total number of PV modules	$N_{tot} = N_s \times N_p$	162.00	—
Maximum power per PV module	$P_{max.module}$	619.9966	W
Cells per module	N_{cell}	156.00	—
Open-circuit voltage per module	V_{oc}	55.55	V
Short-circuit current per module	I_{sc}	14.25	A
Voltage at MPP per module	V_{mp}	45.79	V
Current at MPP per module	I_{mp}	13.54	A
PV array voltage at MPP	$V_{mp,array} = N_s \times V_{mp}$	412.11	V
PV array current at MPP	$I_{mp,array} = N_p \times I_{mp}$	243.72	A
PV array power at MPP	$P_{mp,array}$	100.439	kW
Temperature coefficient of V_{oc}	αV_{oc}	-0.250	%/°C

Temperature coefficient of I_{sc}	α_{Isc}	0.045	%/°C
PV rated power used for design	P_s	100.00	kW
DC-link voltage	V_{dc}	600	V
PV-side current used for Boost design	I_e	278.00	A
DC-link current	I_s	166.00	A
Boost switching frequency	$f_{sw, boost}$	5.00	kHz
Inverter switching frequency	$f_{sw, inv}$	10.00	kHz
Current ripple	ΔI	20.00	A
Voltage ripple	ΔV	6.00	V
Boost inductance	L	1.45	mH
Input capacitor	C_1	1000.00	μF
DC-link capacitor	C_2	3227.70	μF
Series resistance	R_1	1.00	mΩ
Initial MPPT reference voltage	$V_{ref, init}$	300.00	V
Maximum MPPT reference voltage	$V_{ref, max}$	363.00	V
Minimum MPPT reference voltage	$V_{ref, min}$	0.00	V
MPPT voltage step	ΔV_{ref}	1.00	V
Nominal irradiance	G	1000.00	W/m ²
Cell temperature	T	25.00	°C

Note: PV = photovoltaic; MPP = maximum power point; MPPT = maximum power point tracking; “—” indicates not applicable.

Numerical simulations were carried out in the MATLAB/Simulink 2022b environment, selected for its capabilities in multi-domain modeling, control implementation, and dynamic analysis of grid-connected PV systems.

3.2 Integrated Control Strategy

The proposed control methodology follows a hierarchical structure designed to coordinate maximum PV power extraction, DC-link voltage regulation, grid synchronization, and active/reactive power management. In this framework, the MPPT controller governs the Boost converter to maintain the PV generator at its optimal operating condition. Meanwhile, the grid-side subsystem, relying on an SRF-PLL together with synchronous dq-frame vector control, is responsible for stabilizing the DC-link voltage and ensuring the injection of high-quality current into the utility grid. A major benefit of this configuration is the effective separation between the slower dynamics associated with the PV source and the faster dynamics of the grid interface, while still preserving consistent energy transfer throughout the entire conversion process [9].

3.2.1 Development of the maximum power point tracking control strategy

(a) Principle of MPPT control

MPPT is performed using a hybrid P&O–WOA approach that combines the straightforward implementation of the Perturb and Observe technique with the strong global optimization capability of the WOA. The tracking mechanism relies on monitoring variations in the PV output power, expressed as $P_{pv} = V_{pv} \times I_{pv}$, produced by perturbations applied either to the PV voltage or to the converter duty cycle. In the conventional P&O method, when the power variation satisfies $\Delta P_{pv} > 0$, the perturbation direction is maintained, whereas a decrease in power ($\Delta P_{pv} < 0$) requires reversing the perturbation direction. Although this strategy is efficient and easy to implement near the maximum power point, its performance deteriorates when the P–V characteristic contains several local maxima, especially under partial shading conditions. To overcome this drawback, the WOA is incorporated to perform a broader global search capable of identifying the GMPP while avoiding convergence toward local optima. In the proposed framework, the monitored quantity is the PV power P_{pv} , and the resulting output is the optimal duty cycle D applied to the Boost converter [10].

(b) Coupling of MPPT control with the boost converter

The Boost converter acts as the main control element of the MPPT mechanism. Assuming ideal operation in continuous conduction mode, the relationship between the PV voltage and the DC-link voltage can be written as:

$$V_{dc} = \frac{V_{pv}}{1 - D} \quad (1)$$

where, D represents the converter duty cycle. Consequently,

$$D = 1 - \frac{V_{pv}}{V_{dc}} \quad (2)$$

These expressions indicate that any modification in the duty cycle directly influences the voltage applied to the PV array and, therefore, changes its operating point along the I–V curve. The PV output power is defined by:

$$P_{pv} = V_{pv} I_{pv} \quad (3)$$

Accordingly, for a fixed DC-link voltage, changing D alters the PV voltage, V_{pv} , which consequently shifts the operating condition of the PV source on its characteristic curve. The objective of the MPPT controller is therefore to regulate the duty cycle in such a way that V_{pv} remains close to the optimal voltage V_{MPP} . An increase in the duty cycle modifies the electrical loading conditions seen by the PV generator, resulting in a displacement of the operating point toward a different intersection on the equivalent load line associated with the Boost converter input. Hence, the interaction between the MPPT algorithm and the Boost converter forms the fundamental mechanism through which the system controls and optimizes energy extraction from the PV source [12].

(c) MPPT control performance

The hybrid P&O–WOA approach is developed to satisfy four primary goals: rapid convergence toward the optimal operating point, high tracking precision, minimization of oscillations around the maximum power point, and strong robustness under changing irradiance conditions. Within this framework, the WOA contributes an efficient global exploration capability, particularly in nonlinear and multimodal operating regions, while the Perturb and Observe method ensures accurate local adjustment through a computationally simple decision process. The combination of these two techniques is expected to decrease convergence time, reduce steady-state power oscillations, and enhance the overall efficiency of energy extraction. In addition, because the control output is expressed as a duty-cycle reference instead of a directly imposed voltage, the method remains fully compatible with the physical behavior of the DC–DC converter and can be smoothly integrated into the complete PV conversion system [13].

Consistent with the quantitative comparison reported in Section 2.1 (Table 2 and Table 3), the selection of WOA as the global-search component of the hybrid strategy is driven primarily by its superior transient tracking speed and its robustness against local maxima under multimodal or rapidly varying irradiance conditions, rather than by a reduction in computational burden. Under the harmonized computational budget ($N \times \text{Iter}_{\max} = 200$ evaluations/cycle) common to all four bio-inspired hybrids benchmarked, WOA's search mechanism imposes a per-cycle computational load comparable to, or lower than, that of the alternative hybrids considered, so that the observed gains in tracking speed and re-tracking robustness are obtained at no additional computational premium [14].

3.2.2 Phase-locked loop synchronization strategy

The SRF-PLL operates by converting the three-phase voltage signals into a rotating synchronous dq reference frame. When the Park transformation is applied, it yields the following mathematical relations.

$$\begin{bmatrix} V_d \\ V_q \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos \theta & \cos \left(\theta - \frac{2\pi}{3} \right) & \cos \left(\theta + \frac{2\pi}{3} \right) \\ -\sin \theta & -\sin \left(\theta - \frac{2\pi}{3} \right) & -\sin \left(\theta + \frac{2\pi}{3} \right) \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} \quad (4)$$

When the reference frame is perfectly synchronized with the grid voltage, the quadrature component vanishes: $V_q = 0$.

The phase error is then processed through a PI controller:

$$\omega = \omega_0 + K_{p,PLL} V_q + K_{i,PLL} \int V_q dt \quad (5)$$

and the estimated angle is obtained by integration:

$$\hat{\theta} = \int \omega dt \quad (6)$$

The PLL thus provides the angle θ required for the $\frac{abc}{dq}$ and $\frac{dq}{abc}$ transformations.

The PLL does more than just estimate the phase angle; it is a key element in inverter control. Its performance has a direct impact on how well the d- and q-axis components are separated, which in turn influences the precision of power regulation and the quality of the current injected into the grid. If the synchronization is not accurate, it can cause unwanted interaction between the axes, poorer current tracking, and increased harmonic distortion [15].

In this context, the SRF-PLL is a suitable choice for grids that are balanced or only slightly disturbed. This is mainly due to its straightforward implementation, its compatibility with vector control methods, and its ability to maintain stable and reliable dynamic performance.

3.2.3 Active and reactive power control strategy

The control on the grid side is developed in a synchronous dq reference frame that is oriented with the grid voltage. By ignoring the zero-sequence component and assuming that $V_q \approx 0$ thanks to the PLL action, the expressions for active and reactive power can be written as follows:

$$P = \frac{3}{2} (V_d i_d + V_q i_q) \approx \frac{3}{2} V_d i_d \quad (7)$$

$$Q = \frac{3}{2} (V_q i_d - V_d i_q) \approx -\frac{3}{2} V_d i_q \quad (8)$$

These expressions indicate that, when the reference frame is correctly synchronized, the active power is primarily influenced by i_d , whereas the reactive power is governed by i_q . This characteristic supports the application of indirect power control by regulating the current components.

The outer control loop of the DC-link voltage is responsible for producing the reference value of the active current. If we define the voltage error as:

$$e_{vdc} = V_{dc}^* - V_{dc} \quad (9)$$

and the PI control law is expressed as:

$$i_d^* = K_{p,dc} (V_{dc}^* - V_{dc}) + K_{i,dc} \int (V_{dc}^* - V_{dc}) dt \quad (10)$$

Then, we obtain:

$$i_d^* = K_{pv} e_{vdc} + K_{iv} \int e_{vdc} dt \quad (11)$$

This control loop maintains the balance of energy between the power generated on the PV side and the power delivered to the grid.

By considering the decoupling terms that arise from the RL filter model connecting the inverter to the grid, the current dynamics can be described as follows:

$$L \frac{di_d}{dt} = v_{inv,d} - V_d + \omega L i_q - R i_d \quad (12)$$

$$L \frac{di_q}{dt} = v_{inv,q} - V_q - \omega L i_d - R i_q \quad (13)$$

where, v_{id} and v_{iq} are the control voltages generated by the inverter. The current errors are defined as:

$$e_d = i_d^* - i_d \quad (14)$$

$$e_q = i_q^* - i_q \quad (15)$$

The internal PI controllers are given by:

$$u_d = K_{pi} e_d + K_{ii} \int e_d dt \quad (16)$$

$$u_q = K_{pi} e_q + K_{ii} \int e_q dt \quad (17)$$

The control voltages applied to the inverter are then constructed as follows:

$$v_{inv,d}^* = V_d - \omega L i_q + K_{p,d} (i_d^* - i_d) + K_{i,d} \int (i_d^* - i_d) dt \quad (18)$$

$$v_{inv,q}^* = V_q + \omega L i_d + K_{p,q} (i_q^* - i_q) + K_{i,q} \int (i_q^* - i_q) dt \quad (19)$$

In this approach, the reference for the reactive current is chosen as $i_q^* = 0$, which guarantees operation at unity power factor. When the decoupling terms are incorporated, the resulting voltage references applied to the inverter can be expressed as:

$$v_{id}^* = u_d + V_d + Ri_d - \omega Li_q \quad (20)$$

$$v_{iq}^* = u_q + V_q + Ri_q + \omega Li_d \quad (21)$$

These expressions enable fast and decoupled tracking of the i_d and i_q currents.

This reflects a mode of operation where reactive power is effectively zero and the power factor is ideally equal to one. Such an approach focuses on maximizing the delivery of active power from the PV system, while also simplifying the control scheme and keeping the grid current aligned with the voltage. Overall, the control architecture clearly distributes the tasks: the MPPT optimizes energy harvesting, the DC-link voltage loop ensures proper energy exchange with the grid, and the dq current control loops regulate the quality and composition of the injected current [16].

3.2.4 Controller parameterization

(a) WOA parameter settings

The WOA stage searches over the boost converter duty cycle $D \in [0.1, 0.9]$, maximizing the instantaneous PV power $P(D) = V(D) \cdot I(D)$ as its fitness function. The parameter settings, consistent with the companion benchmarking study conducted on the same PV–boost architecture, are summarized in Table 7.

Table 7. WOA parameter settings for the proposed P&O–WOA MPPT algorithm

Parameter	Symbol	Value	Role
Population size	N	10	Number of candidate duty-cycle solutions evaluated per iteration
Maximum iterations per MPPT decision	$\text{Iter}_{\max}$	20 (200 evaluations/cycle)	Search-effort budget, harmonized across all hybrids compared
Convergence parameter	a	Linearly decreased $2 \rightarrow 0$	Governs the shift from exploration to exploitation
Spiral shape constant	b	1	Shape of the spiral bubble-net update
Control-loop sampling time	T_s	1 ms	MPPT decision rate
Stopping criterion	ε	10^{-3}	Relative fitness-improvement threshold over a fixed number of consecutive iterations
Safeguard fallback	—	5 consecutive cycles without improvement	Reverts to plain P&O to prevent numerical stagnation (not a global-convergence guarantee)

Note: WOA = Whale Optimization Algorithm; MPPT = maximum power point tracking; P&O = Perturb and Observe; “—” indicates not applicable.

(b) Cascaded-loop PI tuning: symbolic derivation

The DC-link voltage loop (Eqs. (9)–(11)) and the inner dq current loops (Eqs. (14)–(21)) are tuned through a standard cascaded architecture. Rather than tuning both loops against a single lumped closed-loop specification, the design is carried out in two explicit stages, each justified analytically below: the inner current loops are parameterized first from the linearized grid-coupling model, and the outer DC-link loop is parameterized second from the linearized DC-link energy balance, under a bandwidth-separation condition whose validity is established formally rather than assumed [17].

(b.1) Inner current loop—pole-zero cancellation (internal model control)

From the decoupled current dynamics of Eq. (12) and Eq. (13), each axis of the grid-coupling filter behaves, at the small-signal level, as a first-order RL system relating the control voltage to the corresponding current:

$$G_i(s) = \frac{I(s)}{U(s)} = \frac{1}{Ls + R}$$

where, L and R denote the equivalent inductance and series resistance of the grid-side coupling filter appearing in Eq. (12) and Eq. (13) (distinct from the boost-stage inductance already listed in Table 4). A PI controller is placed in series with this plant:

$$C_i(s) = K_{pi} + \frac{K_{ii}}{s} = K_{pi} \frac{\tau_i s + 1}{\tau_i s}, \quad \tau_i = \frac{K_{pi}}{K_{ii}}$$

Selecting the controller zero to cancel the plant pole, i.e. setting $\tau_i = L/R$, removes the plant dynamics from the open-loop expression:

$$L_i(s) = C_i(s)G_i(s) = K_{pi} \frac{\tau_i s + 1}{\tau_i s} \cdot \frac{1}{Ls + R} = \frac{K_{pi}}{Ls}, \quad \text{for } \tau_i = \frac{L}{R}$$

So that the resulting closed-loop transfer function reduces to a single first-order system whose pole location is fixed entirely by the proportional gain:

$$T_i(s) = \frac{L_i(s)}{1 + L_i(s)} = \frac{\omega_{c,i}}{s + \omega_{c,i}}, \quad \omega_{c,i} = \frac{K_{pi}}{L}$$

This yields the inner-loop design relations, expressed symbolically as functions of the target current-loop bandwidth $\omega_{c,i}$ and the (as yet unspecified) filter parameters L and R :

$$K_{pi} = L\omega_{c,i}, \quad K_{ii} = R\omega_{c,i}$$

Because the pole-zero cancellation eliminates the plant dynamics from the open-loop transfer function, the closed-loop sensitivity and complementary sensitivity functions take the canonical first-order forms

$$S_i(s) = \frac{s}{s + \omega_{c,i}}, \quad T_i(s) = \frac{\omega_{c,i}}{s + \omega_{c,i}}$$

confirming, independently of the numerical values ultimately assigned to L , R and $\omega_{c,i}$, that the closed inner loop is unconditionally stable (single left-half-plane pole) and exhibits the expected high-frequency roll-off of a first-order low-pass system, with no resonant peaking that could otherwise interact adversely with the switching ripple of the boost and inverter stages ($f_{sw,boost}$, $f_{sw,inv}$, Table 6).

(b.2) Outer DC-link voltage loop—linearized energy-balance model

The outer loop regulates the DC-link voltage by shaping the active-current reference i_d^{ref} supplied to the inner loop above. Its plant is obtained not from Eqs. (9)–(11) directly, but from the underlying power balance of the DC-link capacitor, linearized around the nominal operating point (V_{dc0} , Table 6). Neglecting losses and assuming the reactive-current reference is held at zero ($i_q^{\text{ref}} = 0$, unity power factor, consistent with the choice already stated in Section 3.2.3), the instantaneous power delivered to the grid is proportional to i_d , so that the capacitor energy balance

$$C_2 V_{dc} \frac{dV_{dc}}{dt} = P_{pv} - \frac{3}{2} v_d i_d$$

linearizes, for small perturbations ΔV_{dc} and Δi_d around V_{dc0} , into a pure integrator plant:

$$\frac{\Delta V_{dc}(s)}{\Delta i_d(s)} = -\frac{3v_{d0}}{2C_2 V_{dc0} s}$$

where, v_{d0} denotes the nominal d-axis grid-voltage component (a function of the grid phase-voltage amplitude, fixed by the PLL synchronization of Section 3.2.2), and C_2 and V_{dc0} are the DC-link capacitance and operating-point voltage already tabulated in Table 4. A PI controller $C_v(s) = K_{p,dc} + \frac{K_{i,dc}}{s}$ closes this loop; because the plant is a bare integrator rather than a first-order lag, pole-zero cancellation is not applicable here, and the two gains are instead assigned by direct pole placement of the resulting second-order closed loop. The closed-loop characteristic polynomial is:

$$s^2 + \frac{3v_{d0}K_{p,dc}}{2C_2 V_{dc0}} s + \frac{3v_{d0}K_{i,dc}}{2C_2 V_{dc0}} = 0$$

Matching this polynomial term-by-term against the canonical second-order form $s^2 + 2\zeta_v \omega_{n,v} s + \omega_{n,v}^2$ gives the outer-loop design relations, again expressed purely as functions of the desired closed-loop natural frequency $\omega_{n,v}$ and damping ratio ζ_v :

$$K_{p,dc} = \frac{4\zeta_v \omega_{n,v} C_2 V_{dc0}}{3v_{d0}}, \quad K_{i,dc} = \frac{2\omega_{n,v}^2 C_2 V_{dc0}}{3v_{d0}}$$

By the Routh–Hurwitz criterion applied to this second-order polynomial, both coefficients being strictly positive—which holds automatically for any $\zeta_v > 0$ and $\omega_{n,v} > 0$, since C_2 , V_{dc0} , and v_{d0} are all positive physical quantities—is necessary and sufficient for closed-loop stability, so that the outer loop is stable for the entire admissible range of design parameters and does not depend on any particular numerical instantiation [18–20].

3.3 Operating Conditions

The performance of the overall control scheme is examined using two complementary cases. These are designed to highlight both the steady-state behavior of the system and its ability to handle changes in solar input. The first case represents normal operating conditions, while the second introduces variations in irradiance to evaluate how effectively the control loops respond and interact throughout the conversion process.

3.3.1 Steady-state condition

The steady-state analysis is performed under standard conditions, where the irradiance is fixed at 1000 W/m^2 and the cell temperature is kept at 25°C . This case serves as a baseline for assessing system performance. It helps verify that the MPPT algorithm can accurately reach the maximum power point under stable conditions, while also confirming that the DC-link voltage remains properly regulated around its reference value. In addition, it allows evaluation of the quality of the electrical quantities delivered to the grid.

The study mainly examines the average power extracted, the small oscillations around the maximum power point, the stability of the DC-link voltage, and the waveform quality of grid-side currents and voltages. As such, this scenario provides a clear reference for evaluating tracking precision, power factor performance, and harmonic behavior under normal operation [21–24].

3.3.2 Dynamic condition

In the dynamic case, the irradiance is varied over time while keeping the temperature fixed at 25°C . The goal is to investigate how robust the control strategy is when the available solar power changes. This situation is particularly useful for studying the interaction between the MPPT algorithm, the Boost converter, the DC-link voltage control, and the grid-side vector control.

Changes in irradiance directly modify the Power–Voltage characteristics of the PV array, which simultaneously challenges the MPPT tracking accuracy, the DC-link voltage dynamics, and the quality of power injected into the grid.

This analysis makes it possible to measure the system's response speed, observe temporary variations in V_{dc} , ensure continuous power delivery to the grid, and detect any degradation in current quality during transitions. Therefore, this scenario serves as a key benchmark for evaluating the overall reliability and robustness of the proposed control structure [25–27].

3.4 Evaluation Metrics

The system performance is evaluated using a combination of indicators that capture energy efficiency, electrical stability, and the quality of power delivered. A key parameter is the PV power output, defined as $P_{pv} = I_{pv} \times V_{pv}$, which directly represents how effectively the system converts the available solar energy at the source.

This evaluation is further supported by the MPPT efficiency, commonly expressed as the ratio between the power actually extracted and the maximum theoretical power available. This measure provides insight into how accurately the system tracks the optimal operating point, both in steady conditions and during rapid changes:

$$\eta_{\text{MPPT}} = \frac{P_{pv}}{P_{\text{MPP}}} \times 100 \quad (22)$$

Another important aspect is the stability of the DC-link voltage, V_{dc} , which plays a crucial role in linking the DC and AC stages. Its performance is assessed by examining the steady-state deviation from the reference, the level of voltage ripple, and the extent of transient fluctuations around the desired value V_{dc}^* . Stable DC-link behavior indicates proper coordination between the MPPT controller, the Boost converter, and the inverter.

In addition, attention is given to the magnitude and shape of the grid-side voltages and currents. These signals help determine how well the system follows its references, maintains sinusoidal current waveforms, limits distortion, and preserves proper phase alignment between voltage and current. Within the synchronous reference frame aligned with the grid voltage, the exchange of active and reactive power can be described using the following expressions:

$$P = \frac{3}{2} (v_d i_d + v_q i_q) \quad (23)$$

$$Q = \frac{3}{2} (v_q i_d - v_d i_q) \quad (24)$$

which, under the condition $v_q \approx 0$, reduce to:

$$P \approx \frac{3}{2} v_d i_d \quad (25)$$

$$Q \approx -\frac{3}{2} v_d i_q \quad (26)$$

These performance measures help confirm that the control approach successfully enables the transfer of active power to the grid while keeping reactive power at the intended level, which in this case is close to zero to achieve unity power factor.

In addition, the THD of the injected currents is considered a key indicator of power quality. It provides an overall measure of waveform distortion and is determined using the following expression:

$$\text{THD}_i = \frac{\sqrt{\sum_{n=2}^{\infty} I_n^2}}{I_1} \times 100 \quad (27)$$

where, I_1 represents the fundamental component of the current, while I_n corresponds to the harmonic components of order n . The THD serves as an overall indicator that reflects the effectiveness of the output filter, the quality of the modulation process, and the proper tuning of the control loops. Altogether, these performance metrics offer a comprehensive evaluation of the system by linking energy utilization, dynamic behavior, and the quality of power injected into the grid [28–30].

4 Simulation Results and Discussion

This section presents an analysis of the simulation outcomes for the grid-connected PV system, considering two distinct operating scenarios: a steady-state case under standard conditions (1000 W/m², 25 °C), and a dynamic case with varying irradiance while maintaining a constant temperature of 25 °C. These two situations provide a comprehensive basis for evaluating the system's behavior in terms of energy extraction, DC-link voltage stability, grid-side control performance, and the quality of the injected power.

Since the steady-state and dynamic scenarios probe the same control architecture under fundamentally different stress conditions. Table 8 brings their respective indicators face to face—PV tracking, DC-link settling and ripple, THD, duty-cycle behavior, and MPPT efficiency—turning what would otherwise be a scattered, waveform-by-waveform reading into a single point of cross-scenario verification.

Table 8. Consolidated performance indicators—steady-state vs. dynamic operation

Indicator	Steady-State (1000 W/m ² , 25 °C) Section 4.1	Dynamic (1000 → 700 → 400 → 600 → 800 → 1000 W/m ² steps)—Section 4.2
P_{pv} tracking	≈ 100 kW, ± 1–2% ripple around the MPP (Figure 4)	Near-proportional to I_r : ≈ 100/70/40/60/80 kW at 1000/700/400/600/800 W/m ² (Figure 9)
V_{dc} settling time	≈ 0.02 s to the 600 V reference (Figure 6)	Only small transient deviations per irradiance step; no resettling instability (Figure 10)
V_{dc} steady-state ripple/startup overshoot	± 5–10 V (≈ 1.3–1.7%); startup overshoot ≈ 650 V (+8.3%) (Figure 6)	Comparable ripple ; startup overshoot ≈ 650 V only, no reovershoot at later steps (Figure 10)
THD _i	0.15–0.35% in steady state (avg. ≈ 0.25%); startup spike ≈ 1.8% (Figure 5)	0.2–0.35% in steady intervals; transient peaks ≈ 0.7% at each irradiance step (Figure 14)
D	Settles from 0.40 to ≈ 0.30; ripple ±0.005–0.01 (Figure 7)	Settles from 0.40 to ≈ 0.30; minor, well-damped step transients (Figure 11)
η_{MPPT}	98.44% (P&O–WOA) vs. 97.37% (classical P&O), companion algorithm-level benchmark at STC	Not evaluated in the companion benchmark (STC only); reported here for context only

Note: P_{pv} = photovoltaic power; I_r = solar irradiance; V_{dc} = DC-link voltage; THD_i = current total harmonic distortion; D = duty cycle; MPPT = maximum power point tracking; η_{MPPT} = MPPT efficiency; P&O = Perturb and Observe; WOA = Whale Optimization Algorithm; STC = Standard Test Conditions.

4.1 Scenario 1: Steady-State Operation Under Nominal Conditions

This scenario is evaluated under nominal conditions with a constant irradiance of 1000 W/m² and a cell temperature of 25 °C. It allows for an overall assessment of synchronization with the grid, stability of the control loops, regulation of the DC-link voltage, and the effectiveness of the MPPT strategy.

4.1.1 Three-phase voltages

The three-phase voltages, shown in Figure 2, are balanced and exhibit nearly ideal sinusoidal waveforms, with a phase shift of 120° between each phase. Their peak values are close to ± 340 V, which corresponds to a root mean square (RMS) value of approximately 240 V per phase. This regular and symmetrical behavior confirms that the system is properly synchronized with the grid, and that the PLL ensures a stable rotating reference frame essential for accurate dq decoupling.

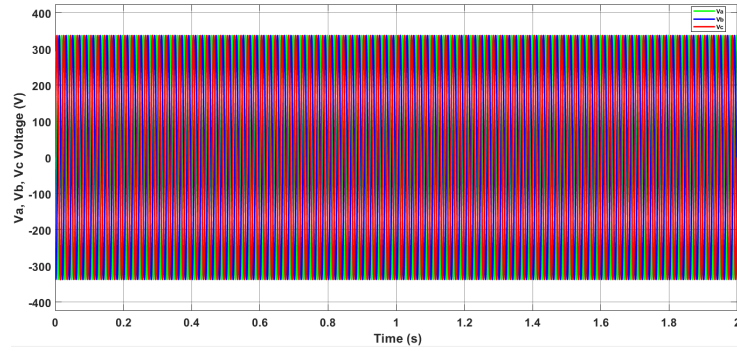


Figure 2. Filter output voltage under steady-state conditions

Note: V_a , V_b , and V_c denote the three-phase voltages.

Beyond simply meeting waveform requirements, these observations indicate that the interaction between the inverter, filter, and grid remains stable under nominal conditions. There are no noticeable signs of imbalance, voltage drops, or phase irregularities. Such performance reflects a well-tuned control system, as any issues with synchronization or controller adjustment would typically appear as distortions or asymmetries in the voltages and currents delivered to the grid.

4.1.2 Injected currents

Under steady-state operation, the injected currents shown in Figure 3 reach peak values of around ± 220 A and maintain balanced, nearly sinusoidal waveforms with the expected phase separation between phases. This behavior indicates precise tracking of the current references and a stable transfer of power, consistent with operation at a power factor very close to unity. In addition, the time-domain response shows no noticeable overshoot or phase mismatch, which reflects the strong dynamic stability of the system. Overall, these observations demonstrate the reliability of the control strategy and its ability to maintain accurate and well-regulated current injection under steady-state conditions.

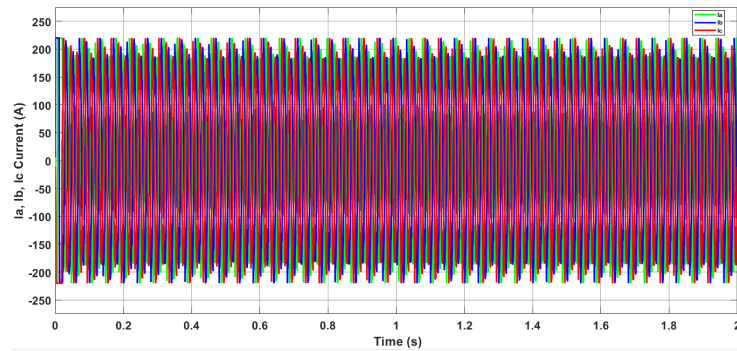


Figure 3. Waveforms of the three-phase currents after filtering under steady-state conditions

Note: I_a , I_b , and I_c denote the three-phase injected currents after filtering.

4.1.3 Photovoltaic power

The PV output power shown in Figure 4 quickly settles at a stable level close to 100 kW, with only minor fluctuations of about 1–2% around the average value. This behavior demonstrates that the MPPT algorithm is able to keep the system operating near the maximum power point with good accuracy under nominal conditions. During startup, a short transient phase is observed, characterized by an initial drop of roughly 270 kW followed by an overshoot reaching nearly 120 kW. However, this response is rapidly attenuated and does not persist. Despite this relatively energetic initial behavior, the system quickly reaches stability, indicating both fast convergence and

effective power tracking. The small oscillations observed in steady state are consistent with the typical behavior of perturb-and-observe or hybrid MPPT techniques. Rather than indicating a limitation, they reflect the usual compromise between responsiveness, implementation simplicity, and precision in tracking the maximum available energy.

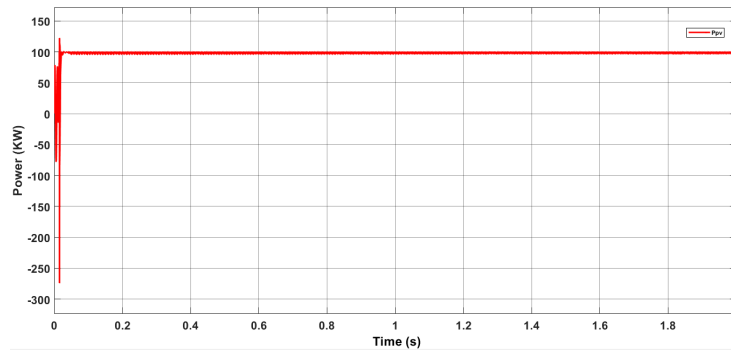


Figure 4. Time evolution of the photovoltaic power (P_{pv}) under steady-state conditions

These qualitative observations are quantitatively confirmed in Table 8: the steady-state P_{pv} stabilizes at approximately 100 kW with only $\pm 1\text{--}2\%$ ripple, consistent with the MPPT efficiency of 98.44% achieved by the proposed P&O–WOA scheme versus 97.37% for classical P&O (companion algorithm-level benchmark. Section 2.1).

4.1.4 Total harmonic distortion

The THD shown in Figure 5 initially shows a significant transient spike, reaching a value close to 1.8%, before quickly settling into a steady operating range. In steady state, it remains confined within a relatively narrow band, typically between 0.15% and 0.35%, with an average value around 0.25. This pattern indicates that harmonic disturbances are mainly present during the startup phase, while the system maintains low and stable distortion once normal operation is achieved. Overall, these findings highlight the effectiveness of the proposed control scheme in limiting harmonic content, demonstrating its robustness and its ability to ensure good power quality under steady operating conditions.

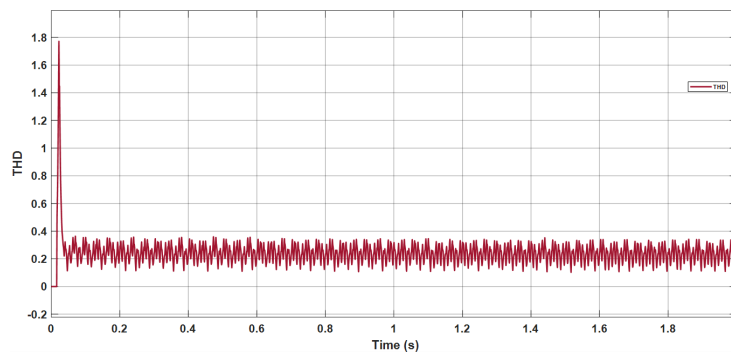


Figure 5. Time evolution of total harmonic distortion (THD) under steady-state conditions

As consolidated in Table 8, this corresponds to a steady-state THD_i confined between 0.15% and 0.35% (average $\approx 0.25\%$), with harmonic content essentially limited to a transient startup spike of $\approx 1.8\%$.

4.1.5 DC-link voltage

The DC-link voltage shown in Figure 6 smoothly settles at its reference value of 600 V within a short response time of about 0.02 seconds. In steady state, it exhibits only a small ripple, typically within ± 5 to ± 10 V, which indicates effective and stable regulation. At startup, a limited overshoot reaching approximately 650 V is observed, but it is short-lived and quickly attenuated without affecting the system's overall stability. These results confirm solid DC-link voltage control, although slight improvements in transient performance could further enhance the behavior during the initial phase.

Table 8 summarizes these values numerically: a settling time of ≈ 0.02 s to the 600 V reference, a steady-state ripple of $\pm 5\text{--}10$ V ($\approx 1.3\text{--}1.7\%$), and a startup overshoot of ≈ 650 V ($+8.3\%$).

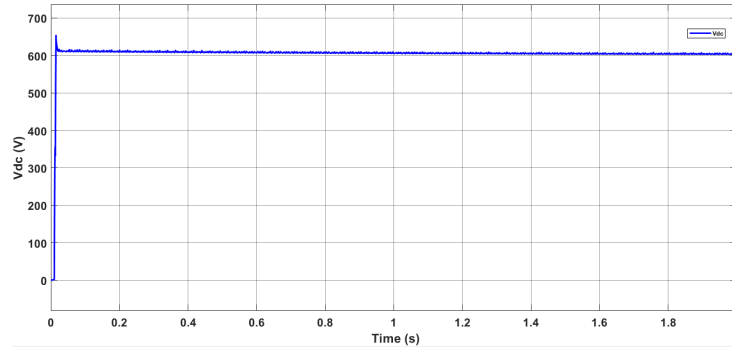


Figure 6. DC-link voltage (V_{dc}) under steady-state conditions

4.1.6 Duty cycle

The duty cycle shown in Figure 7 quickly stabilizes, decreasing from an initial value of 0.40 to an average level close to 0.30. This behavior indicates that the Boost converter successfully reaches a steady operating point that supports optimal power extraction. In steady-state conditions, the variations in the duty cycle remain minimal, typically within a range of about ± 0.005 to ± 0.01 . This low and consistent ripple reflects a well-stabilized MPPT process. Although the initial transient is relatively fast, it remains properly damped, allowing a smooth transition toward stable operation. However, further refinement during startup could help reduce the temporary stresses observed during system initialization.

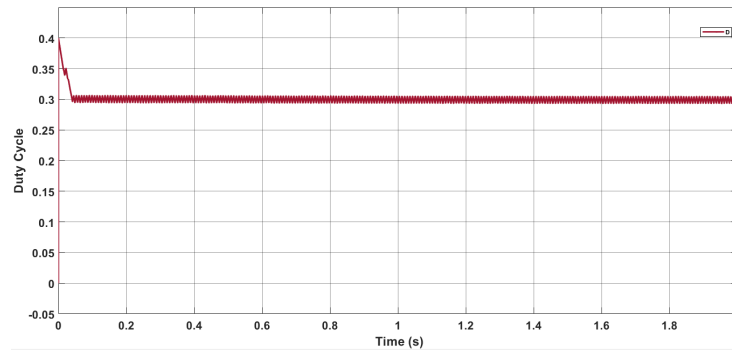


Figure 7. Time evolution of the duty cycle (D) under steady-state conditions

4.2 Scenario 2: Dynamic Operation Under Variable Irradiance

This scenario is designed to examine how the grid-connected PV system responds to changes in solar input occurring in successive steps. The cell temperature is maintained at 25 °C throughout the simulation, while the irradiance is varied to evaluate the system response under dynamic operating conditions. It provides a comprehensive view of the system's behavior under dynamic conditions, enabling the assessment of MPPT responsiveness, DC-link voltage control, stability at the grid interface, and the quality of the power delivered when operating away from steady-state conditions.

4.2.1 Solar irradiance profile

The irradiance profile shown in Figure 8 is varied in a stepwise manner, taking successive values of 1000, 700, 400, 600, 800, and back to 1000 W/m^2 , followed by a similar pattern of decrease and increase over the simulation period. This variation creates a meaningful dynamic input, allowing the evaluation of how well the control strategy can respond to sudden changes in available solar power while preserving overall system stability.

The sequence of clearly defined irradiance levels makes it possible to closely observe the transient behavior of the MPPT and the different control loops at each transition point. As a result, this scenario is particularly suitable for analyzing the balance between rapid adaptation, DC-link voltage stability, and the continuity of power delivery to the grid.

4.2.2 Photovoltaic power

The PV power shown in Figure 9 closely tracks the changes in irradiance, reaching steady levels of about 100, 70, 40, 60, and 80 kW for irradiance values of 1000, 700, 400, 600, and 800 W/m^2 , respectively. This almost

proportional relationship demonstrates the effectiveness of the MPPT in keeping the system operating near its optimal point, even under varying conditions. Although the initial transient response is more noticeable, it remains short-lived and quickly settles. Transitions between different irradiance levels are handled smoothly, without causing instability. Overall, these results confirm that the system achieves reliable and stable energy extraction despite fluctuations in solar input.

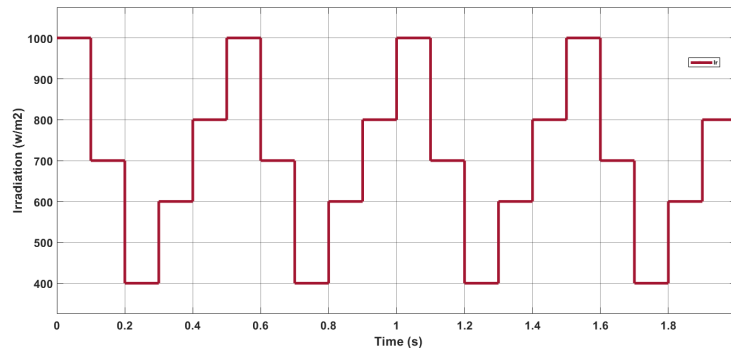


Figure 8. Time evolution of solar irradiance (I_r) under dynamic conditions

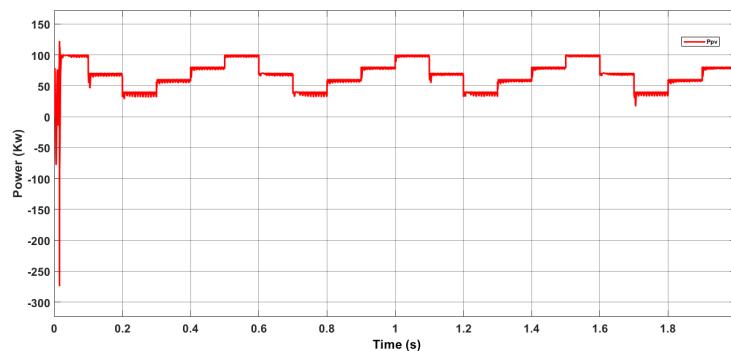


Figure 9. Time evolution of the photovoltaic power (P_{pv}) under dynamic conditions

These results are quantified in Table 8, which reports near-proportional P_{pv} values of approximately 100, 70, 40, 60, and 80 kW for irradiance levels of 1000, 700, 400, 600, and 800 W/m^2 , respectively.

4.2.3 DC-link voltage

The DC-link voltage shown in Figure 10 remains well controlled around its reference value of 600 V, even when the irradiance changes. Only small transient deviations are observed, along with a low ripple once steady state is reached. This behavior demonstrates the effectiveness of the DC-link control loop in maintaining a proper balance between the power drawn from the PV source and the power delivered to the grid. During startup, a limited overshoot of about 650 V occurs, but it is short in duration and quickly settles without affecting system stability. Overall, these results confirm the robustness of the DC-link regulation under varying conditions, although further refinement could help improve the transient response during the initial phase.

As detailed in Table 8, the ripple magnitude remains comparable to the steady-state case, and the startup overshoot (≈ 650 V) is not repeated at subsequent irradiance steps, confirming the absence of re-settling instability.

4.2.4 Duty cycle

The duty cycle shown in Figure 11 quickly settles from an initial value of 0.40 to an average level of about 0.30, after which it exhibits only minor oscillations in steady-state operation. During changes in irradiance, the system undergoes small and well-controlled transient adjustments, demonstrating the Boost converter's ability to adapt effectively while keeping the operating point close to the maximum power point. The minimal residual ripple in the duty cycle further indicates that the MPPT control is stable and well damped, supporting efficient energy extraction even under varying conditions.

This behavior is consistent with the stable MPPT efficiency and low steady-state ripple reported in Table 8, confirming that duty-cycle regulation underpins the quantitative tracking performance summarized therein.

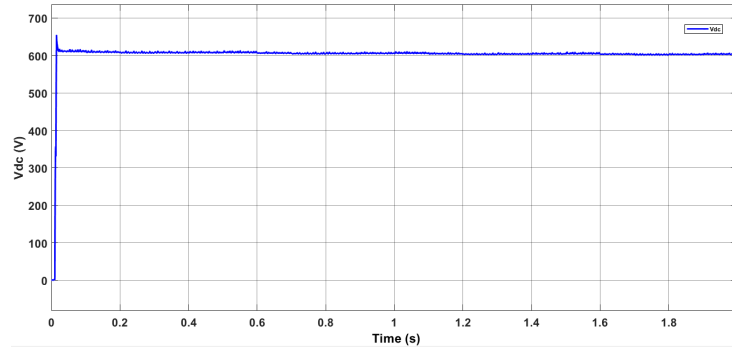


Figure 10. DC-link voltage (V_{dc}) under dynamic conditions

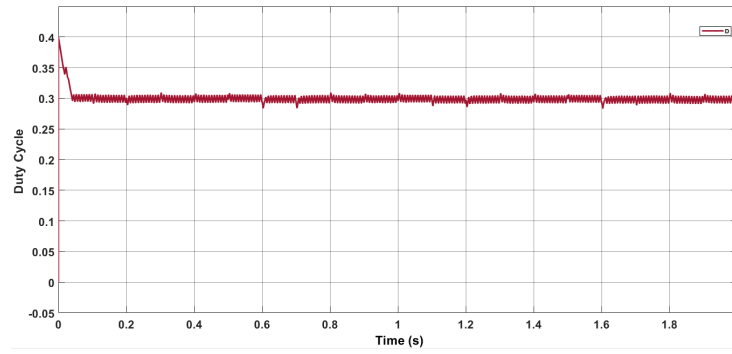


Figure 11. Time evolution of the duty cycle (D) under dynamic conditions

4.2.5 Three-phase voltages

The three-phase voltages shown in Figure 12 remain balanced and sinusoidal, maintaining a consistent phase shift of 120° and a peak amplitude close to ± 340 V throughout the simulation. The lack of noticeable distortion, even with fluctuations in PV power, confirms that the grid voltage is properly enforced and that the PLL ensures stable synchronization. This observation is significant, as it shows that variations on the source side do not disturb the integrity of the grid interface, meaning that the dynamics of the PV generator and the Boost converter are effectively decoupled from the voltage reference imposed by the grid.

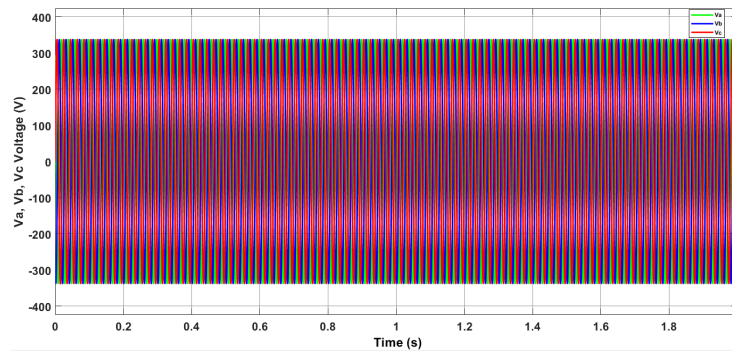


Figure 12. Voltage after filtering under dynamic conditions

Note: V_a , V_b , and V_c denote the three-phase voltages.

This grid-side stability is achieved despite the P_{pv} variations quantified in Table 8, confirming the decoupling between source-side dynamics and grid voltage regulation.

4.2.6 Injected currents

Under dynamic conditions, the injected currents shown in Figure 13 adjust effectively to changes in irradiance by varying their amplitude while preserving a balanced three-phase structure and maintaining near-sinusoidal waveforms. The current magnitude evolves smoothly, with peak values ranging roughly between ± 100 A and ± 220 A depending on the available power, demonstrating the capability of the control system to regulate active power

injection without compromising waveform quality. Additionally, transitions between different irradiance levels occur without noticeable instability or significant phase mismatch. Overall, these observations confirm the strong dynamic performance of the system and highlight the effectiveness of the control strategy in ensuring reliable current tracking under variable operating conditions.

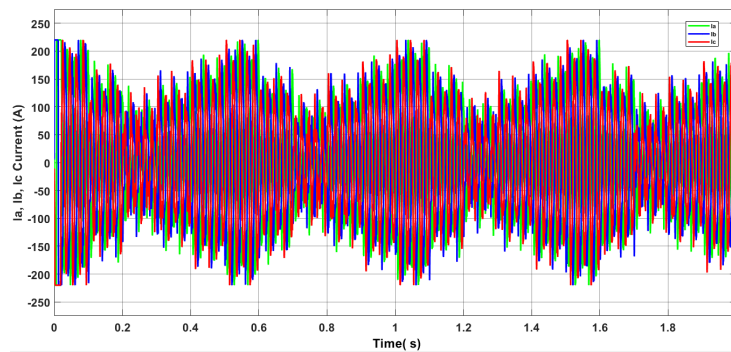


Figure 13. Currents after filtering under dynamic conditions

Note: I_a , I_b , and I_c denote the three-phase injected currents after filtering.

The corresponding current quality is quantified in Table 8 through the THD_i values reported for dynamic operation (0.2–0.35% in steady intervals, transient peaks $\approx 0.7\%$), confirming that current tracking is achieved without compromising harmonic performance.

4.2.7 Total harmonic distortion

The THD shown in Figure 14 initially shows a notable transient peak before quickly decreasing to an average normalized value of around 0.25. During dynamic operation, it generally stays within a range of 0.2 to 0.35, with only brief rises up to approximately 0.7 during irradiance changes. This indicates that harmonic disturbances are mainly associated with transition periods, while outside these intervals the quality of the injected current remains stable and within acceptable limits. Overall, these results highlight the strong harmonic resilience of the proposed system and demonstrate the effectiveness of the control strategy in maintaining good power quality despite varying operating conditions.

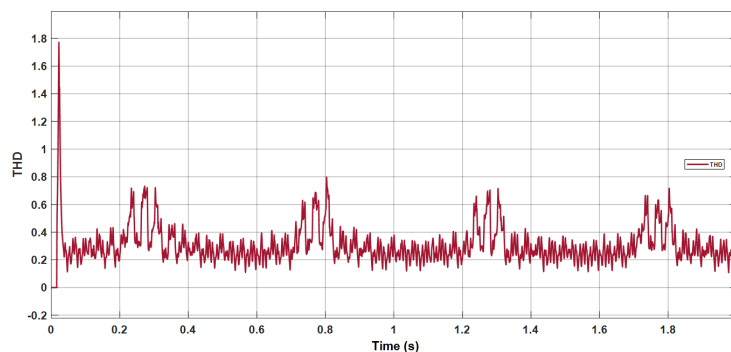


Figure 14. Time evolution of total harmonic distortion (THD) under dynamic conditions

These values are consolidated in Table 8: THD_i remains within 0.2–0.35% during steady intervals, with transient peaks reaching approximately 0.7% at irradiance transitions.

4.3 Synthesis

Table 8 consolidates the quantitative performance indicators discussed in Sections 4.1 and 4.2 across both operating conditions, confirming an MPPT efficiency of 98.44%, a DC-link settling time of approximately 0.02 s with ± 5 –10 V ripple, and a current THD consistently below 0.35% outside transient intervals. Building on these figures, the following discussion interprets the underlying transient mechanisms.

At startup, the system undergoes a noticeable transient phase, characterized by fluctuations in the PV power P_{pv} , a short-term overshoot in the DC-link voltage V_{dc} , and a temporary increase in harmonic distortion of the injected current. This behavior arises from the interaction between the MPPT–Boost conversion stage, the DC-link voltage control, and the grid-side current regulation. In fact, the power generated by the PV source can be expressed as follows:

$$P_{pv}(t) = V_{pv}(t)I_{pv}(t) \quad (28)$$

Whereas, for an ideal Boost converter operating in continuous conduction mode, the generator voltage satisfies:

$$V_{pv}(t) = V_{dc}(t) [1 - D(t)] \quad (29)$$

Thus, any rapid variation in the duty cycle D instantaneously shifts the operating point of the PV generator, directly affecting the extracted power.

Moreover, the energy balance of the DC link can be expressed as:

$$\frac{1}{2}C_{dc} \frac{dV_{dc}^2}{dt} = P_{pv} - P_{ac} - P_{loss} \quad (30)$$

or, equivalently,

$$C_{dc}V_{dc} \frac{dV_{dc}}{dt} = P_{pv} - P_{ac} - P_{loss} \quad (31)$$

where, P_{ac} represents the power delivered to the grid, while P_{loss} accounts for the total system losses. These relationships indicate that, during startup, any temporary imbalance between the power extracted from the PV source and the power supplied to the grid directly leads to variations in the DC-link voltage, which explains the observed overshoot in V_{dc} . However, this effect should be seen as a normal transient phenomenon rather than a fundamental limitation, since it is short-lived and quickly attenuates. Once this initial phase passes, the system settles into stable operation, with efficient power extraction ($P_{pv} \approx 100$ kW), well-regulated DC-link voltage ($V_{dc} \approx 600$ V), balanced three-phase voltages, nearly sinusoidal currents, and low harmonic distortion. Overall, despite some room for improvement during startup, the results demonstrate the strong dynamic performance and effectiveness of the proposed control approach [31–35].

These conclusions are directly supported by the quantitative indicators consolidated in Table 8, rather than by waveform observation alone.

4.4 Main Limitations of the Study

This study has a few limitations that deserve consideration. To begin with, the validation relies entirely on simulation results, without experimental verification. As a result, certain practical aspects—such as hardware imperfections, real-time implementation constraints, and parameter uncertainties—are not fully captured. In addition, the analysis is based on simplified assumptions that are necessary to make the model manageable, but these may reduce how closely the results reflect real operating conditions.

Furthermore, some important scenarios have not been explored. In particular, the irradiance profiles considered in Section 3.3 are spatially uniform: the steady-state case (1000 W/m²) and the stepwise dynamic case (1000→700→400→600→800→1000 W/m²) probe DC-link/PLL/current-loop coordination under source-side transients, but do not reproduce spatially non-uniform, multi-peak conditions such as partial shading, nor grid-side disturbances such as voltage sags, unbalanced faults, or frequency excursions. Detailed thermal effects on the PV panels and the integration of energy storage systems were likewise not investigated. The reported results should therefore not be interpreted as evidence of performance under partial-shading or grid-disturbance conditions. Consequently, the findings should be viewed as a coherent numerical validation of the proposed approach under the two tested scenarios, rather than a complete proof of its performance under all possible operating conditions [36–39].

Beyond these general limitations, three practical factors specific to translating the proposed hybrid MPPT and control structure into a digital implementation deserve explicit mention. First, although the WOA stage operates under a fixed, harmonized computational budget ($N \times \text{Iter}_{\max} = 200$ duty-cycle evaluations per MPPT decision) whose per-iteration cost, $O(N)$, is no higher than that of the P&O–GWO hybrid and lower than that of P&O–BAT or P&O–SFLA, and this load falls within the range commonly reported as feasible on TI C2000- or STM32-class DSP/FPGA controllers at the millisecond-range sampling rate used in this study ($T_s = 1$ ms), this remains an implication drawn from operation counts and reported feasibility ranges rather than a measured execution time on target hardware. Second, a digital realization of the proposed framework would need to address fixed-point quantization of the duty-cycle search variable and of the PI integrator states, and the synchronization of ADC sampling with the PWM update instant, both of which can introduce bias and delay not present in the continuous-time simulation. Third, sensor noise and measurement delay on the voltage and current feedback used by the P&O–WOA fitness evaluation and by the dq current loops could, in principle, affect MPPT convergence and current-loop accuracy in ways not captured by the present noise-free model. These factors are not modeled in the present study and are identified here as explicit targets for experimental verification [40–46].

4.5 Future Research Directions

Future developments can be outlined along seven main directions. First, spatially non-uniform irradiance conditions, including partial shading and multi-peak P–V profiles, should be investigated to assess the GMPP-tracking capability of the P&O–WOA scheme beyond the globally-uniform irradiance ramps and steps considered in the present study. Second, grid-side disturbance scenarios—voltage sags, unbalanced faults, and frequency excursions—should be introduced to evaluate the resilience of the PLL synchronization and current-control loops under abnormal grid conditions. Third, experimental implementation should be carried out to validate the proposed architecture in real-world conditions and confirm the practical robustness of the MPPT, PLL, and grid-side control, including hardware-in-the-loop (HIL) or prototype testing on a representative DSP/FPGA target (e.g., TI C2000- or STM32-class), measurement of the actual worst-case execution time of the P&O–WOA routine, and assessment of fixed-point quantization, ADC/PWM synchronization, and sensor-noise effects identified in Section 4.4.

Fourth, incorporating advanced control methods, such as robust or predictive strategies, could further improve system performance in the presence of uncertainties and grid disturbances. Fifth, particular attention could be given to improving startup behavior, especially by introducing precharging techniques and better coordination between control loops during transients. Sixth, enhancing DC-link voltage regulation would help reduce transient deviations and strengthen the interaction between the MPPT stage and the inverter. Finally, extending the proposed system to microgrid applications and advanced reactive power management represents a promising direction, enabling the PV inverter to actively support voltage regulation and provide additional grid services.

5 Conclusion

This work has introduced a comprehensive framework for the control and conversion of a grid-connected PV system. The approach is based on the coordinated integration of a hybrid P&O–WOA MPPT algorithm, a Boost converter, an SRF-PLL, and an indirect P/Q control strategy implemented in the synchronous dq reference frame. The main contribution lies in combining energy extraction, DC-link regulation, grid synchronization, and inverter-side control within a unified and consistent structure.

Simulation results demonstrate that, under nominal conditions, the system reaches a stable operating point, with PV power close to 100 kW, a well-regulated DC-link voltage around 600 V, balanced three-phase voltages, nearly sinusoidal injected currents, and acceptable harmonic performance. Under dynamic conditions with varying irradiance, the system maintains strong adaptability: the extracted power follows the imposed levels accurately, the DC link remains stable, and the quality of power injection is effectively preserved.

Author Contributions

Conceptualization, W.O.A. and N.B.; methodology, W.O.A. and N.B.; software, W.O.A.; validation, W.O.A., N.B., F.A.K., and M.G.; formal analysis, W.O.A. and N.B.; investigation, W.O.A.; resources, N.B. and M.G.; data curation, W.O.A.; writing, original draft preparation, W.O.A.; writing, review and editing, N.B., F.A.K., and M.G.; visualization, W.O.A.; supervision, N.B. and M.G.; project administration, N.B.; funding acquisition, N.B. and F.A.K. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

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